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The History of Music

Approaches from Cultural Evolution and Musicology

C8S1 *Mason Youngblood and Fabian C. Moss*

Introduction

C8P1 Explicit references to music are common in Darwin's early writings on biological evolution. At the same time, music historians draw on analogies of natural growth to describe music's changing styles. In an article considered the foundational document for musicology as an academic discipline, Austrian musicologist Guido Adler (1855–1941) sketches the history of music as a quasi-evolutionary process. For him, studying the history of music entails comparative research on forms of music notation and writing systems, the development of classification taxonomies for different musical forms, styles, and genres, as well as “the investigation of the laws of art of different periods” (Mugglestone & Adler, 1981). These “laws of art,” he explains further, can be understood in analogy to biological growth, as the central concern for the new field of historical musicology “is to demonstrate and establish how, proceeding from the beginnings of simple melody, the structure of works of art gradually grows” (Mugglestone & Adler, 1981). Setting the progressionist undertones aside, the influences of Darwin's revolutionary conceptions of biological evolution on Adler's thinking about music history and culture are evident. Since then, however, music research has fragmented along methodological and institutional lines, with historical musicology and systematic musicology (e.g., psychology,

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acoustics) under the roofs of the humanities and the sciences, respectively. The resulting gap prevents true interdisciplinary exchange to the present day. While the biological evolution of musicality has remained fairly active in recent decades (see other contributions in this volume), quantitative research on music history has just recently been revitalized by the emerging field of cultural evolution, which has the potential to bridge this gap.

C8P2 Below, we summarize a wealth of recent research on the cultural evolution of music in broad strokes, focusing on studies on the contents and contexts of music as well as novel methodologies. After that, we identify five remaining issues and open questions. In particular, we briefly discuss epistemological differences, issues of data preservation, debates surrounding innovation in aesthetic settings, and the challenges posed by globalization and technological advancements, and we briefly elaborate on the tension between conceptions of function and progress. These five issues and questions are not meant to be the most important to either historical musicology or cultural evolution—instead, they represent overlapping interests of the two fields. We conclude with areas where we see promising potential for future interactions between historical musicology and cultural evolution. Note that this chapter reviews only a selection of research at the junction of historical musicology and cultural evolution, as we are unable to provide more than a handful of citations here because of word limits—more comprehensive reference lists can be found in Savage (2019) and Youngblood et al. (2023).

C8S2 Cultural Evolutionary Research on Music

C8P3 A major focus of cultural evolution research on music has been describing and explaining cross-cultural similarities in musical features and structure. Central findings include that scales in traditional music from around the world tend to be composed of five or seven equal steps, and rhythms usually exhibit isochrony with smaller subdivisions of two or three beats (Savage et al., 2015). These

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regularities are driven, at least in part, by cognitive biases for simplicity and predictability. In iterated learning experiments where participants transmit information to one another along a chain, similar to the children's game "telephone," melodies and rhythms converge toward cross-cultural universals while displaying variation that reflects local music traditions (Anglada-Tort et al., 2023). Studies also show that large-scale trends in popular music may be shaped by cognitive biases—for novelty and rarity in the pop charts, negative emotion in lyrics, and conformity in the samples that artists use (Youngblood, 2019).

C8P4 Context and community structure also play a major role in shaping musical diversity. Large-scale cross-cultural studies indicate that the context in which music is produced heavily predicts its acoustic features (e.g., lullabies and love songs have a lower tempo than dance songs) (Mehr et al., 2019), so much so that people are able to reliably assign songs from different cultures to the context in which they appear (Yurdum et al., 2023). For instance, group singing is much more common than solo singing cross-culturally, especially in contexts such as dance, religion, and mourning. Most work of what we know about community structure comes from contemporary music, where detailed data on artist relationships are easily accessible. Cultural artifacts like music samples appear to "diffuse" through networks of collaborating artists, which loosely map on to genres and subgenres. Genetic population structure, on the other hand, does not have a consistent relationship with traditional musical variation. At global scales there does not appear to be a strong correlation between genetic and musical variation (Passmore et al., 2024).

C8P5 One of the biggest contributions of cultural evolution to the study of music are new quantitative methods for description, some of which are based on explicit analogies with biological evolution. Researchers have applied these tools to reassess historical findings and subject them to empirical testing. Phylogenetic reconstruction, originally designed to infer species relationships from genetic data or physical traits, has been adapted to investigate plausible lineages of manuscript copies, the historical trajectories of melodies and songs, modes, rhythms, and instruments. Sequence alignment,

which aligns DNA sequences with mutational differences, has been repurposed to track changes in English and Japanese folk music melodies. More broadly, statistical modeling approaches from ecology and the social sciences have allowed researchers to track the historical development of harmony in Western classical music (Moss et al., 2024) as well as major trends in popular music including lyrics, harmonics, timbre, and pitch and chord transitions (Mauch et al., 2015).

C8S3 **Bridging Epistemological Differences**

C8P6 While cultural evolution and historical musicology share a common interest in understanding how music developed in the past, and how music cultures, styles, and technologies have changed over time, they often disagree with respect to particular research interests and epistemologies. Musicologists, and humanities scholars more generally, are often interested in “thick description,” the minute study of (ideally) all relevant details and contextual aspects, and express skepticism toward generalizing statements and findings that claim to explain historical processes, which is a common goal in the social sciences, including cultural evolution. Interdisciplinary tensions are thus no surprise. However, as both perspectives embody the desire to reduce uncertainty surrounding knowledge of the past, we believe that these can be at least partially reconciled if there is a deeper conversation about epistemic goals.

C8P7 In particular, the act of describing something in detail can be understood as a form of data modeling. The development and quality assessment of descriptive models, such as standardized encoding formats for sheet music or the reconstruction of historical networks, then, adhere to similar scientific criteria as those for predictive or causal models. For instance, explicitly stating model assumptions, avoiding under- or overfitting, and seriously engaging in model comparison may build conceptual bridges.

C8P8 Additionally, historians have begun to more seriously engage with “counterfactual histories” as a research methodology. In

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simple terms, this is an attempt to play through plausible alternative scenarios to real historical events, constrained by what is already known. Far from being mere “fan fiction,” these accounts aim at better understanding what was in light of what could have been. This loudly resonates with empirical simulation of sampling-based studies, such as prior/posterior predictive analyses in the context of Bayesian analytical frameworks.

C8P9 Cross-fertilizing the rigorous statistical methodology from the social sciences with the acute awareness for cultural intricacies of the humanities seems a most promising avenue for historical musicology and cultural evolution to interact. That being said, different epistemologies are not only possible but desirable if they are complementary or, at least, provoke productive interdisciplinary exchanges. A multifaceted phenomenon like music necessitates and deserves a wide range of scientific approaches. What would be beneficial both to the humanistic and scientific study of music is to identify specific areas of tension and to collaborate on their resolution.

C8S4 **Handling Biased Historical Preservation**

C8P10 Apart from institutional obstacles, interdisciplinary research on the history of music that applies both quantitative and qualitative methods is also limited by biases in the historical record. The myriad ways of preserving and transmitting music between individuals and from one generation to the next range from explicit oral one-on-one instruction to implicit imitation to different forms of writing systems and codifications. As with any cultural artifact, the written record in stone, paper, or silicon usually facilitates the reconstruction of things of the past, whereas preservation is limited to the memory of living individuals and societies for oral cultures. Moreover, different notational systems omit different musical facets that are not deemed important or assumed to be known implicitly by anyone. Selection biases become amplified even more strongly when some traditions are valued more than others. The music of certain individuals or groups may be banned or celebrated; judgments of

differing aesthetic or economic value may lead to conservation or canonization of some but not of others.

C8P11 Most important, even when forms of notation exist, they are not the music itself but an attempt to write down what is essentially ephemeral. Music historiographies, in particular in the West, have focused mostly on its notational and representational aspects, as those can be read from the written record. But, as uses of music change, for example, responding to larger sociopolitical developments or simply to the dynamics of taste and fashion, so do its functions. While drawing inferences about past musical behaviors from implicit notational forms (e.g., in the forms of instruments or drawings or dance) is challenging, it is clear that research into the history and cultural evolution of music needs to be based on a larger basis than notated musical pieces alone. In short, there is no archeological record of music, only of music codifications. Studying music history quantitatively requires sophisticated models that do not hide these issues but rather express the associated uncertainty.

C8P12 Cultural evolution and musicology can also collaborate and reinforce one another in the areas of cultural heritage preservation. Coordinated efforts to document endangered music cultures threatened by globalization or climate change are needed in order to safeguard humanity's musical heritage, and provide "missing links" in the evolutionary history of human music-making (Wallin et al., 2001).

C8S5 Effects of Globalization and Technology

C8P13 One area where musicology could inform research on the cultural evolution of music is the role of globalization and technology in musical change. For decades, musicologists have been deeply concerned about the effects of global interconnectedness on cultural diversity. Alan Lomax may put it most strongly: "A mismanaged, over-centralized electronic communication system is imposing a few standardized, mass-produced and cheapened cultures everywhere" (Lomax, 1977). The limited quantitative research on music

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globalization broadly supports this assertion. European pop music charts have become increasingly American-dominated since the 1960s, and releases from countries with more cultural influence are more likely to become hits. At the same time, the spread of music production technologies can lead to surprising and fascinating new forms of musical variation. Auto-tune, for example, has become extremely popular among singers in North Africa because of how it interacts with melismatic vocals (Clayton, 2016). Some of this variation traces colonial and postcolonial human migration. Sound system culture—innovated in Jamaica and brought to the UK by the Windrush generation—laid the groundwork for half a century of popular music in the UK, from post-punk to jungle and drum and bass.

C8P14 Cultural evolution is well equipped to study how increases in global connectivity and the diffusion of new technologies affect music. Network analysis, acoustic feature analysis, and other methods have been used to address a variety of related questions, such as the role of the internet in delocalizing artist communities and the impact of wars on music discovery behavior. However, sensitive studies of globalization and technology diffusion require the kind of in-depth domain knowledge that historical musicologists are able to contribute. Work like this also often requires big data, and at this point most datasets on contemporary music are Western-centric and rather small. Luckily, recent years have seen the development of new corpora that focus on music outside of the traditional Western canon. The Global Jukebox, which expands on Alan Lomax's groundbreaking Cantometrics project from the 1980s, includes original recordings with standardized acoustic features from over 1,000 societies (Passmore et al., 2024). Another notable example is the CompMusic project, which includes thousands of transcriptions and recordings of music from India (Carnatic and Hindustani), North Africa (Arab-Andalusian), Turkey (*makam*), and China (*jingju*). Both historical musicology and cultural evolution are themselves undergoing a process of “globalization” to consider the full diversity of human cultural life (Kroier, 2012). Interdisciplinary

collaborations should catalyze this process by prioritizing the analysis and construction of cross-cultural datasets.

C8S6 **Disentangling Evolution and Progress**

C8P15 Unfortunately, teleological misconceptions of evolution are commonplace. The “march of progress” cartoon—depicting a linear transition from apes to humans—continues to shape popular perceptions about evolution, despite the fact that biologists have ejected the progressivist view of evolution for decades. In the words of Stephen Jay Gould, progress is a “noxious, culturally embedded, untestable, nonoperational, intractable idea that must be replaced if we wish to understand the patterns of history” (Gould, 1988). An early example for a progressionist model of music evolution is given by the Belgian music theorist François-Joseph Fétis (1784–1871). According to his theory, music is governed by a system of relations between tones termed “tonality,” a neologism at the time. His innovation is to recognize that the organization of tones is subject to geographical and diachronic change and not, as earlier theorists would have it, determined by universal laws (physical or cosmological-divine). Fétis proposes a four-stage model (Fétis, 1844) in which music proceeds from simple monophonic melodies in a single mode or key to a stage where modulations are possible to a stage where multiple keys are equally important to a stage where, finally, all tones and relations between them are equal. His merit is to recognize change in tonal organization, his failure is to assume progress toward an entropic totality. Naturally, applications of cultural evolution to music today are frequently subject to criticisms based on the misconception that evolution implies progress (Savage, 2019).

C8P16 In cultural evolution research, progress is most often discussed in relation to cumulative cultural evolution (CCE)—the improved functionality of behaviors, tools, or other variants over time (Mesoudi & Thornton, 2018). CCE does not imply inevitable “progress” in human culture as a whole, as there are many examples of

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behaviors becoming less complex, efficient, or functional. It simply provides a framework for understanding the “ratchet effect” that occurs in some forms of human culture, like technology. As others have noted, there are several problems associated with applying CCE to aesthetic domains like music. Even if one takes for granted that music has a universal function, such as social bonding, the features associated with that function are contextual and constantly changing. In other words, music and other forms of aesthetic culture are optimized for moving targets. Is it possible to determine whether UK rave music is better at enhancing social bonds than the all-night song and dance traditions of Xingu people? Probably not.

C8S7 **The Generation of New Innovation**

C8P17 Another area where interdisciplinary collaboration between historical musicology and cultural evolution could be particularly fruitful is the question of innovation, which is understudied relative to other drivers of cultural change. Research on the cultural evolution of contemporary music has yielded some interesting insights into how innovations emerge and influence future musical variation. Songs with novel sounds and lyrics do better in the pop music charts, and “underground” artists who introduce novel sounds are often copied by more popular artists. Musical innovations can open up new styles of music, and the originators of these styles experience elevated success over time. That being said, we still lack an understanding of how the creative practices of artists lead to innovations in the first place.

C8P18 Moving toward a better understanding of innovation requires two things: new quantitative models and rich historical examples. Cognitive scientists and information theoreticians have developed several candidate models of innovation that deserve more attention by cultural evolution researchers. Historical musicologists, particularly organologists, have detailed many examples of innovation in instrument designs, notation, and other musical practices that would be excellent models for cultural evolutionary research.

However, truly understanding innovation as a process requires artist- or community-centered examples that include qualitative details about the technologies used, the cultural context, and ideally a record of the artists' creative process (e.g., from journals, letters, or interviews) (Meyer, 1983).

C8P19 Another example might be innovations in tuning systems and temperaments. A number of particular temperaments and the musical affordances they entail have been developed several times, and are extremely well documented by historical and ethnomusicologists. Direct results from innovations in temperament are creative choices such as “triadic atonality”—the juxtaposition of tonally distant harmonies—featuring centrally in both fifteenth-century motets and nineteenth-century symphonic poems. For the former, it seems that it was not fit enough for the cultural environment at the time (i.e., tuning systems, listeners' taste), whereas the latter flourished and still heavily impacts on contemporaneous film music. Indeed, integrating experimentation, innovation, and creativity into theories of the cultural evolution of music may be crucial to reconcile them with these sorts of historical and cross-cultural accounts.

C8S8 Conclusion

C8P20 The field of cultural evolution has the potential to bridge the gap between historical and systematic approaches to musicology by contributing new methods, models, and cross-cultural insights. In turn, expertise and experience from historical musicology have the potential to illuminate fundamental questions about cultural evolution, such as how and why creative innovations emerge. At least some of the epistemological differences between the two approaches are complementary, and potentially reconcilable. By identifying general trends and cross-cultural mechanisms of musical change, cultural evolutionary researchers can provide a “backdrop” for the more detailed and nuanced accounts of historical musicologists. Model-based simulations may also be useful for addressing other interests

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of humanities scholars, such as dealing with historical uncertainty and exploring counterfactual scenarios.

C8P21 The models and methods used to study the cultural evolution of music are often strikingly similar, if not the same, as those used to study change in other domains like literature and the visual arts. Within cultural evolution, this means that many researchers who study music are also deeply interested in other forms of aesthetic culture. There is also a great deal of overlap with the computational humanities and empirical aesthetics communities, with members of all three fields attending similar conferences, co-hosting workshops, and so forth. As a result, we feel that cultural evolution may act as an interdisciplinary umbrella connecting different areas of interest within the humanities, and linking them to psychological and cognitive work being done on music, art, and aesthetics.

C8P22 In this chapter, we have focused on high-level ways that interdisciplinary collaboration can contribute to our understanding of music history, but we will close on a more practical note—with recommendations for how cultural evolution researchers can directly engage with musicology. First, the historical musicology literature is full of interesting, untested hypotheses that could be investigated using quantitative methods. Cultural evolutionary researchers are also often interested in whether models are able to generalize to unseen or unusual data, and there is rich documentation of such cases by musicologists. Second, cultural evolution usually analyzes artifacts, whereas musicologists do often focus on practices and experiences. The latter are less reductive, and capture more of the complexity of real musical behavior. Finally, one pitfall of cultural evolution is that methods sometimes take precedence over questions. In other words, researchers sometimes find datasets and questions that they can apply their existing methods to, rather than prioritizing the questions and developing new models to address them. Musicology is much more focused on building domain knowledge—something that we think cultural evolution could benefit from.

Overarching Research Questions

- C8P23 • How can cultural evolution and historical musicology reconcile their epistemological differences?
- C8P24 • How do we address the data quality issues introduced by biased preservation?
- C8P25 • What roles do globalization and technology play in the cultural evolution of music?
- C8P26 • How can we investigate the functions of music while avoiding concepts of progress?
- C8P27 • How and why are new innovations generated in music?

Take-Home Messages

- C8P28 • Music from around the world often exhibits common features, such as scales with five or seven equal steps and isochronous rhythms.
- C8P29 • Cross-cultural similarities in traditional music, as well as contemporary trends in popular music, appear to be driven by biological and cognitive biases for various features.
- C8P30 • Acoustic features are strongly associated with the context in which music is produced, and people globally can reliably pair songs to their context based on these features.
- C8P31 • Community structure (e.g., who performs with or learns from whom) plays a major role in shaping musical diversity, whereas genetic population structure does not seem strongly correlated with musical variation.
- C8P32 • Cultural evolution has contributed new quantitative methods for description—such as phylogenetic reconstruction, network analysis, and statistical modeling—that have led to significant findings about the history of music.

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